Ambiguity, Investor Composition, and Sovereign Rollover Crises

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Abstract

A rollover crisis depends on the information creditors use and on the creditors who hold the maturing debt. This paper places heterogeneous doubts about the bias of official information in a global game of sovereign rollover. A creditor with ambiguity radius δ_k acts as if the official signal were lower by δ_k. The run cutoff rises by $(\alpha/\beta)\delta_k$, where α and β are public and private signal precision. The selected equilibrium remains unique under the usual precision bound. Investor composition then moves a unique crisis threshold through the run propensities of the creditor types. Greater official precision stabilizes after favorable news and destabilizes after unfavorable news. Greater credibility lowers the threshold at every announcement. Max-min ambiguity is observationally equivalent to a Bayesian belief in a pessimistic signal bias, so spreads do not identify ambiguity preferences. Revision histories and individual forecasts must discipline the bias set before market prices are used to evaluate the coordination mechanism. A documented scenario produces ambiguity-equivalent spread increments of 18 basis points in calm conditions and 98 basis points in stress. These magnitudes are model scenarios, not estimates of an ambiguity premium.

JEL codes: D81, D82, F34, G01, H63.
Keywords: rollover risk, global games, ambiguity, multiple priors, investor base, transparency.

1 Introduction

In October 2009 the incoming Greek government announced that the budget deficit was far larger than previously reported. The estimate was revised again. The European Commission's January 2010 report documented serious weaknesses in Greek fiscal statistics (European Commission, 2010). The episode changed the information problem facing a creditor. A published fiscal number could no longer be treated as an unbiased signal with ordinary sampling error.

This paper studies that doubt and the investor base that carries it. Creditors face an unknown bias in official information and evaluate rollover under the worst prior in an interval, following Gilboa and Schmeidler (1989). Creditor types may differ in their rollover and exit payoffs. The investor base gives the fraction of maturing debt held by each type. The model asks how these objects enter the crisis threshold.

The model has an exact as-if representation. Type k behaves like a Bayesian creditor who lowers the official signal by δ_k. Its private-signal cutoff rises by $(\alpha/\beta)\delta_k$. Distrust is costly where creditors must place greater posterior weight on official information. The representation also establishes an identification limit. The equilibrium cannot distinguish max-min ambiguity from a Bayesian belief that the signal has a pessimistic bias. Market prices alone cannot identify ambiguity preferences.

The global-game precision bound continues to select a unique equilibrium (Hellwig, 2002; Morris and Shin, 2003). Distrust moves each type's cutoff without changing the slope bound. Investor composition then moves a unique threshold. The derivative is the difference between the types' run probabilities, multiplied by the coordination term $1/(1-\Phi')$. The same mechanism makes a given composition shift more costly when the run margin is densely populated.

Official precision and official credibility are different objects. Precision places more weight on the announcement. It lowers the crisis threshold only when the announcement is sufficiently favorable. Credibility reduces δ_k and lowers the threshold at every announcement. The model gives the exact news threshold at which the sign of precision changes. Ex ante welfare requires integration over the announcements induced by a precision policy and subtraction of its resource cost. A conditional threshold result does not supply an unconditional welfare ranking.

The quantitative exercise reports scenarios. With the payoff and information inputs listed in Table 1, the ambiguity-equivalent spread increment is 18 basis points in calm news and 98 basis points in stress. Raising the fragile share from 0.35 to 0.65 produces a 10 basis point gap in calm conditions and a 44 basis point gap after adverse news. Raising recovery from 0.60 to 0.70 lowers the stress spread from 575 to 128 basis points. These calculations describe the model at stated inputs. They are not estimates of the share of observed spreads caused by ambiguity.

Global games supply the equilibrium machinery (Hellwig, 2002; Morris and Shin, 2003; Morris and Shin, 2004). Multiple-priors preferences follow Gilboa and Schmeidler (1989). Ambiguous information has been studied in asset pricing and macroeconomics (Epstein and Schneider, 2008; Ilut and Schneider, 2014). Ui (2025) studies strategic ambiguity in symmetric global games. Sovereign rollover models provide the debt setting (Calvo, 1988; Cole and Kehoe, 2000; Bocola and Dovis, 2019). The present model connects heterogeneous effective pessimism to investor composition and keeps the selected threshold unique. Holder-sector data then measure the composition state for the historical periods they cover (Arslanalp and Tsuda, 2014).

The empirical sequence respects the equivalence result. Real-time revision histories discipline bounds on δ_k before spreads enter the analysis. Individual forecasts discipline private-signal dispersion and the response to official releases. Current holder-sector accounts measure composition. Spreads and auction outcomes then test the interaction implied by the model. Institutional changes can provide event variation, but adoption and publication timing are selected. The design therefore states the conditions under which those events can be interpreted causally.

2 The Rollover Game

The analytical structure is exact inside the Gaussian one-shot game. Propositions 2 through 6 follow from normal signals, max-min preferences over a bias interval, Assumption 1, and the uniqueness inequality. Section 6 attaches magnitudes to those objects with documented payoffs, investor-base shares, information precisions, and revision-based distrust measures. Section 5 chains the static game into the policy experiments.

2.1 Environment

A sovereign has a unit continuum of short-term creditors, each holding one maturing bond. Each creditor chooses to roll over or to run. The sovereign survives the rollover iff the mass of runners is at most its liquid capacity θ: fundamentals are the fraction of maturing debt the sovereign could redeem using buffers and primary surplus. If $\theta < 0$ the sovereign enters crisis regardless, and if $\theta > 1$ it survives even a full run. Between, survival is a coordination outcome.

Payoffs may differ across creditor types. Rolling over gives R_k if the sovereign survives and $\mathcal{R}_k$ in crisis. Exiting gives Q_k in survival and q_k in crisis. Assume $R_k \geq \mathcal{R}_k$, $Q_k \geq q_k$, and

$$R_k - \mathcal{R}_k > Q_k - q_k, \tag{1}$$

so rollover is more exposed to the crisis state than exit. Type k rolls over when its assessed crisis probability satisfies

$$(1-P)R_k + P\mathcal{R}_k \geq (1-P)Q_k + Pq_k \quad \iff \quad P \leq \bar{p}_k \equiv \frac{R_k - Q_k}{(R_k - \mathcal{R}_k) - (Q_k - q_k)}. \tag{2}$$

The safe-exit benchmark sets $Q_k = q_k = 1 - \kappa_k$. The quantitative section further imposes common payoffs, giving $\bar{p} = 27.3$ percent.

Information has a public and a private layer. All creditors see the official signal $y = \theta + \sigma_y \eta$, precision $\alpha = \sigma_y^{-2}$: the budget documents and the program reviews. Each creditor i also has private information $x_i = \theta + \sigma_x \varepsilon_i$, precision $\beta = \sigma_x^{-2}$, with ε_i independent standard normal: his own analysis and client flows.

The equilibrium formulas use the diffuse-prior limit standard in the Gaussian global game. Conditional on (x_i, y), posterior precision is $\alpha + \beta$. A sequence of proper normal priors with variance tending to infinity gives the same formulas. Ex ante welfare is evaluated under a proper objective distribution in Section 4.1.

2.2 Distrust as multiple priors

The departure is that creditors discount the official signal. Type-k creditors entertain the set of models in which the official signal carries an unknown bias b,

$$y = \theta + b + \sigma_y \eta, \qquad b \in [-\delta_k,\, \delta_k],$$

and evaluate actions by the worst model in the set (Gilboa and Schmeidler, 1989): the max-min creditor asks what the rollover is worth if the books are cooked against him by as much as his doubt allows. The radius δ_k is the paper's measure of distrust: zero for a creditor who accepts official statistics, large after an episode like Greece's 2009 revisions. It is a primitive about the credibility of the *information source*, rather than about risk appetite, and Section 7 ties it to observable revision histories.

The relevant restriction concerns relative state exposure rather than a perfectly safe exit.

Assumption 1 (Common worst-case ordering)**.** *Conditional payoffs* $(R_k, \mathcal{R}_k, Q_k, q_k)$ *do not depend directly on the signal-bias model. Survival weakly raises the payoff to either action. Inequality* (1) *holds and* $0 < \bar{p}_k < 1$*.*

Both actions can now have state-dependent payoffs. A larger perceived crisis probability lowers the value of rollover more than the value of exit, so the same pessimistic model governs the comparison.

Lemma 1 (Worst-case selection and as-if pessimism)**.** *Under Assumption 1, the crisis event in a threshold equilibrium is* $\{\theta \leq \theta^*\}$*. The worst-case model for both actions is* $b = \delta_k$*. Type* k *behaves as a Bayesian creditor who observed* $y - \delta_k$ *and uses the type-specific hurdle* $\bar{p}_k$*.*

Proposition 1 (Observational equivalence)**.** *Conditional on* $(\delta_k, \bar{p}_k)$*, creditor choices, the run mass, the crisis threshold, and model-implied spreads are identical under max-min ambiguity and under a Bayesian model in which type* k *believes the official signal has pessimistic bias* δ_k*. These outcomes cannot distinguish ambiguity aversion from pessimistic bias beliefs.*

Revision histories and institutional changes must therefore identify the admissible bias set before spreads are used to evaluate the coordination mechanism. The model's distinct observable is heterogeneous response to a common announcement once payoff hurdles and prior bias beliefs are separately disciplined.

2.3 The ambiguity premium and the survival of uniqueness

Bayesian updating from the effective signal gives type k's posterior $\theta \mid x_i \sim N(m_k(x_i), (\alpha+\beta)^{-1})$. Threshold strategies are cutoffs x_k^*. The indifference condition $\mathbb{P}(\theta \leq \theta^* \mid x_k^*) = \bar{p}_k$ gives

$$x_k^* = x_{B,k}^*(\theta^*) + \frac{\alpha}{\beta}\delta_k, \tag{3}$$

where $x_{B,k}^*$ contains the type-specific quantile $\Phi^{-1}(\bar{p}_k)$.

Proposition 2 (The ambiguity premium and its loading)**.** *Type* k*'s run cutoff exceeds its Bayesian cutoff by* $(\alpha/\beta)\delta_k$*. This loading is independent of* $\bar{p}_k$*, though the level of the cutoff is not.*

The loading α/β carries the economics. Distrust of the official source matters only insofar as the posterior leans on that source: where private information is abundant (β large), creditors cross-check the government and δ is nearly free. Where the official account is all there is, every unit of doubt translates into cutoff. This is a cross-country prediction, stated as P2 in Section 7, and it inverts a comfortable intuition: the sovereigns most exposed to a credibility loss are precisely those whose information environment official sources dominate.

Aggregating, the mass of runners at fundamental θ is $\sum_k \mu_k \Phi(\sqrt{\beta}(x_k^* - \theta))$ for composition weights μ_k, and the crisis threshold solves the fixed point

$$\theta^* = \sum_k \mu_k \Phi\left(g_k(\theta^*) + \frac{\alpha}{\sqrt{\beta}}\delta_k\right), \qquad g_k(\theta^*) \equiv \frac{\alpha(\theta^* - y) - \Phi^{-1}(\bar{p}_k)\sqrt{\alpha+\beta}}{\sqrt{\beta}}. \tag{4}$$

Proposition 3 (Uniqueness is ambiguity-proof)**.** *If $\alpha/\sqrt{\beta} < \sqrt{2\pi}$, equation (4) has a unique solution, and the threshold equilibrium is the unique equilibrium in monotone strategies, for every composition (μ_k) and every profile of ambiguity radii (δ_k). Ambiguity shifts the map while leaving its slope controlled by the same precision ratio.*

The condition is the familiar one (Hellwig, 2002; Morris and Shin, 2003). Its economic content is independence from δ. Distrust operates like a family of parallel signal translations, so heterogeneous radii preserve the slope condition. The comparative statics are well defined inside the uniqueness region. The scenario ratio is 1.35 against the bound 2.51. Forecast microdata can discipline β, while real-time official forecast errors can discipline α. As the frontier approaches, the multiplier $1/(1-\Phi')$ grows.

On a twelve-cell sensitivity grid, private noise ranges from 0.10 to 0.30 and public noise from 0.25 to 0.50. The ratio $\alpha/\sqrt{\beta}$ runs from 0.40 to 4.80, with 3 cells outside the sufficient uniqueness region. At baseline public noise, the frontier for private noise is 0.279. The ambiguity-equivalent stress increment exceeds the calm increment in every unique cell. The ratio ranges from 1.3 to an unbounded value when the calm increment approaches zero. The grid holds the signal levels fixed and reports scenario sensitivity rather than sampling uncertainty.

Remark 1 (Polarization is informative but not unique)**.** *By Lemma 1, one announcement y enters type k's posterior as $y - \delta_k$. Holding private signals fixed, posterior means differ by $\alpha/(\alpha+\beta)$ times the cross-sectional dispersion of δ_k. A common increase in signal noise does not produce this cross-sectional wedge. Heterogeneous Bayesian bias beliefs can produce the same wedge, as Proposition 1 requires. Survey disagreement around releases therefore disciplines the distribution of effective pessimism. It does not identify ambiguity preferences by itself.*

What the static game leaves open is who the types are and how many of each hold the debt. That composition is the paper's state variable, and the next section makes it the comparative-static of interest.

3 The Investor Base

For exposition, group creditors into stable and fragile types with weights $1-\mu$ and μ. Their payoff hurdles $\bar{p}_S$ and $\bar{p}_F$ may differ. Stable holders have radius δ_S, while fragile holders have

$\delta_F \geq \delta_S$. The quantitative benchmark sets common payoffs and $\delta_S = 0$ to isolate distrust. Empirical implementation retains type-specific hurdles because regulatory treatment, funding needs, and recovery exposure vary across holder sectors.

Proposition 4 (Composition is priced, with amplification and interaction)**.** *At the unique equilibrium of* (4)*, define* $A_k = g_k + (\alpha/\sqrt{\beta})\delta_k$*. The composition derivative is*

$$\frac{d\theta^*}{d\mu} = \frac{\Phi(A_F) - \Phi(A_S)}{1 - \Phi'}.$$

Here $\phi_k = \phi(A_k)$ *and* $\Phi' = (\alpha/\sqrt{\beta}) \sum_k \mu_k \phi_k$*. It is positive if and only if the fragile type is more likely to run at the threshold. Holding payoff hurdles fixed, a rise in* δ_F *raises the crisis threshold. The size of either effect varies with the normal densities at the two run margins and with the coordination multiplier.*

In the common-payoff scenario, the ambiguity-equivalent premium is 18 basis points in calm news and 98 basis points in stress. These are the 18 and 98 basis point scenarios requested by the calibration audit. They are not estimates of an ambiguity premium because Proposition 1 gives the same prices under pessimistic bias beliefs.

The numerator is the direct difference in run propensities. The equilibrium consequence is larger by $1/(1 - \Phi')$. Forecast microdata discipline the direct response and the signal precisions. The fixed-point derivative then determines the implied amplification within the model.

4 Official Information: Precision and Credibility

An official communication policy in this model is a pair (α, δ): how sharp the signal is, and how far it can be doubted. The pair moves the crisis threshold through different doors, and the distinction carries all of the section's content.

Proposition 5 (Precision stabilizes conditionally, credibility unconditionally)**.** *(i) The effect of official precision on the crisis threshold satisfies*

$$\operatorname{sign}\left(\frac{d\theta^*}{d\alpha}\right) = \operatorname{sign}\left(y^\dagger - y\right), \qquad y^\dagger = \theta^* + \frac{\sum_k \mu_k \phi_k \left[\delta_k - \Phi^{-1}(\bar{p}_k)/(2\sqrt{\alpha + \beta})\right]}{\sum_k \mu_k \phi_k},$$

so sharper official information lowers the crisis threshold if and only if $y > y^\dagger$*. The hurdle exceeds* θ^* *when every type has* $\bar{p}_k < 1/2$*. A rise in a type's distrust raises this hurdle in proportion to its mass and its density at the run margin. (ii) Credibility has an unambiguous threshold effect,* $d\theta^*/d\delta_k = \mu_k \phi_k (\alpha/\sqrt{\beta})/(1 - \Phi') > 0$*, for every news level and every composition whenever* $\alpha > 0$*.*

Precision raises the weight placed on the announcement and on the common component of beliefs. It helps after sufficiently favorable news and hurts after sufficiently unfavorable news. Distrust raises the amount of good news required for precision to help. A reduction in δ_k lowers the threshold at every announcement. At the benchmark, halving the fragile type's radius lowers the stress spread by 50 basis points. This is a conditional statement about crisis incidence. It is not yet a welfare ranking.

4.1 Ex ante welfare of official precision

Let $L(\theta) > 0$ denote the social loss from a rollover crisis and let $C(\alpha)$ denote the resource cost of producing a signal with precision α. For a proper prior F and the signal density generated by that prior, ex ante welfare is

$$W(\alpha) = -\int L(\theta)\mathbf{1}\{\theta \leq \theta^*(y, \alpha)\} f_\alpha(y \mid \theta)\, dy\, dF(\theta) - C(\alpha). \tag{5}$$

The derivative of (5) contains the movement of the equilibrium boundary, the change in the distribution of announcements, and the marginal production cost. Proposition 5 signs only the first object conditional on y. It follows that no unconditional welfare ranking of precision follows from the threshold comparative static. A precision program is desirable when the reduction in expected crisis losses, integrated over the announcements it produces, exceeds its cost. A credibility improvement that lowers each δ_k weakly lowers the crisis region pointwise in y. Its gross crisis-loss effect is therefore positive. The net welfare effect still subtracts the cost of the institutional reform. This distinction is the relevant welfare discipline in Angeletos and Pavan (2007).

The next result separates distrust from a second-moment measure.

Proposition 6 (Distrust is directional, noise is anchored)**.** *(i) $d\theta^*/d\delta_k > 0$ whenever type k has positive mass and $\alpha > 0$. (ii) As official precision vanishes, the threshold converges to*

$$\theta^*_\infty = \sum_k \mu_k(1 - \bar{p}_k).$$

The complete removal of public precision changes the threshold by $\theta^*_\infty - \theta^*$. It is destabilizing when the current threshold is below the anchor and stabilizing when the threshold is above it. (iii) In the latter states, the effect of removing public information and the effect of greater distrust have opposite signs. A variance measure is therefore not a global proxy for distrust.

Figure 3 reports a finite scenario in which the official signal standard deviation rises by 0.08. The distrust shift raises the threshold throughout the plotted news range. The noise experiment changes sign because it moves the threshold toward the no-information anchor. This plotted comparison is a scenario result. The proposition signs the distrust derivative and the limit as public precision vanishes.

5 Dynamics and Policy

The dynamic block links successive rollover games through composition, distrust, and fundamentals news. The calculations below change one slow state at a time.

Quantitative tightening as a composition treatment. A central bank holds maturing debt with zero run margin: it is the limiting stable holder. Balance-sheet runoff hands each maturing bond back to the private margin, mechanically raising μ_t. The experiment in Section 6 drifts μ from 0.35 to 0.65 over eight years, the order of magnitude of post-2022 runoff programs relative to maturing stocks, and prices the drift twice: against calm news throughout, and against a moderate deterioration arriving in year seven. The composition premium is 10 basis points in the first case and 44 in the second. QT's rollover cost is a state-contingent exposure, essentially all of which is collected in bad states. A debt manager who evaluates runoff at calm-market spreads is marking an option position at its premium rather than its payoff.

The doom loop. Higher spreads worsen future fundamentals through debt service, which feeds back into y_{t+1}. Appendix C shows the feedback adds a genuinely dynamic instability: past a spread level, the stable fixed point disappears and the sovereign spirals, the Calvo (1988) mechanism in modern dress, quantified for the Bayesian case by Bocola and Dovis (2019). The headline experiments set this feedback to zero to isolate the composition and credibility mechanisms. With a positive feedback, the same wedges shrink the distance between the calm equilibrium and the spiral's basin boundary.

Collective action clauses as coordination policy. CACs and their statutory relatives raise expected recovery $\mathcal{R}$ in restructuring. Through (2) they raise the hurdle $\bar{p}$, making creditors tolerant of more crisis risk before running, and through Proposition 6(ii) they also lower the uninformative-signal anchor $1-\bar{p}$. Both margins are quantified in Section 6: recovery of 0.70 rather than 0.60 cuts the stress spread from 575 to 128 basis points, and shrinks the ambiguity premium itself, from 0.036 to 0.033 in threshold units, because bounding the loss bounds what the worst case can threaten. That last effect is the model's precise version of the proposal's conjecture that contractual design matters more in ambiguous environments: design that compresses worst-case payoffs is a substitute for trust.

Backstops as ambiguity floors. An official facility that commits to lend against fundamentals above some floor truncates the support of the worst case: a creditor may doubt the budget numbers, while the facility itself anchors the lower tail. In multiple-priors terms the backstop cuts δ directly, which by Proposition 5(ii) stabilizes unconditionally, and its value is largest exactly when δ is high, a reading of catalytic finance (Morris and Shin, 2006) in which the catalyst works on the prior set rather than the signal. The corollary for program design is uncomfortable but useful: a backstop whose own activation is discretionary re-introduces the ambiguity it was meant to remove, and prices accordingly.

Table 1: Quantitative scenario.

Parameter		Value	Category	Basis
Rollover yield	R	1.04	scenario	promised gross return
Recovery	$\mathcal{R}$	0.60	external	haircut evidence (Cruces and Trebesch, 2013)
Exit cost	κ	0.08	scenario	exit payoff
Hurdle	$\bar{p}$	27.3%	internal	equation (2)
Private noise	σ_x	0.15	scenario	forecast-microdata target
Official noise	σ_y	0.33	scenario	real-time-error target
Precision ratio	$\alpha/\sqrt{\beta}$	1.35	internal	uniqueness bound 2.51
Distrust radius	δ	0.08	scenario	revision-quantile target
Fragile share	μ	0.40	scenario	current holder-sector target
News, calm / stress	y	1.10 / 0.95	scenario	capacity units

Table 2: Ambiguity-equivalent spread increments.

	Calm ($y = 1.10$)		Stress ($y = 0.95$)	
	Bayesian	Distrust	Bayesian	Distrust
Crisis threshold θ^*	0.377	0.412	0.549	0.584
Crisis probability (%)	1.5	2.0	11.5	13.6
Spread (bp)	61	79	477	575
Ambiguity premium (bp)	18		98	

Notes: $\mu = 0.40$ and $\delta = 0.08$. The multiplier $1/(1 - \Phi')$ is 2.1 in both regimes. "Bayesian" sets $\delta = 0$. By Proposition 1, the same differences arise from Bayesian pessimistic-bias beliefs.

6 Quantitative Magnitudes

6.1 Calibration

Table 1 defines a transparent quantitative scenario. Recovery of 0.60 is an external estimate consistent with the average haircut evidence in Cruces and Trebesch (2013). The promised return and exit cost are payoff scenarios. They imply the rollover hurdle. The signal variances, distrust radius, investor share, and news levels are scenario inputs. Section 7 specifies how forecast microdata, revision histories, and current holder-sector accounts would estimate them without using spreads. The scenario satisfies the uniqueness condition. Its purpose is to measure the nonlinear amplification generated by the model at stated inputs.

6.2 Results

Table 2 reports the model arithmetic. The same effective pessimism raises spreads by 18 basis points in calm news and 98 basis points in stress. The difference arises from the density at the run margin. Figure 1 traces thresholds and stress spreads across investor composition. Figure 2 gives the conditional sign boundary for precision. Figure 3 compares distrust with a finite noise scenario. Figure 4 reports the composition path under stated news.

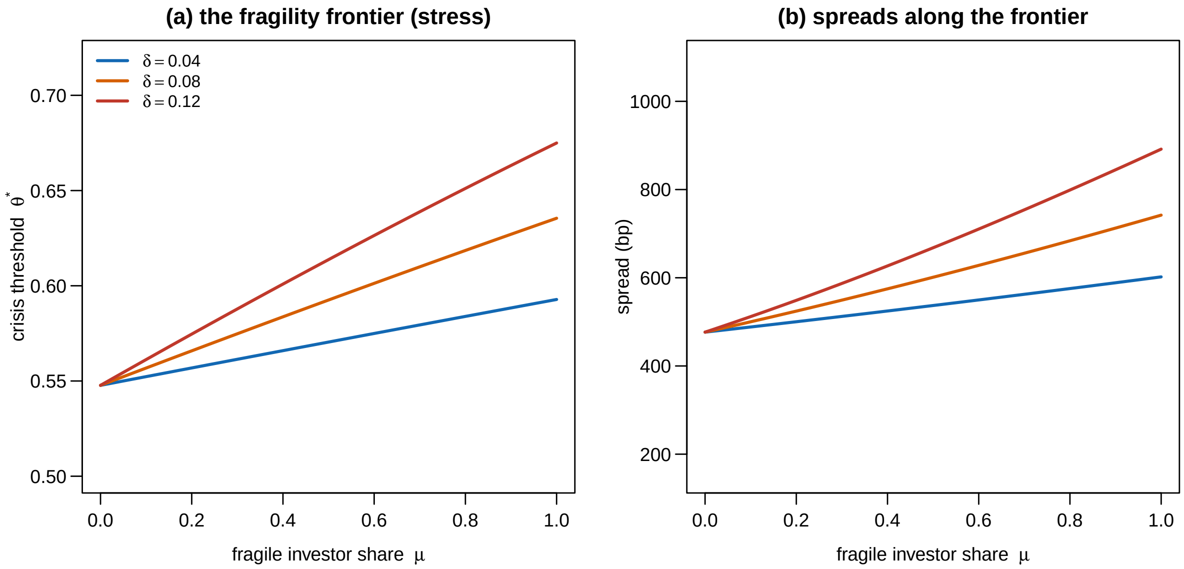

Figure 1: The fragility frontier under stress news: crisis threshold (a) and spread (b) by fragile investor share, for distrust radii $\delta \in \{0.04, 0.08, 0.12\}$.

7 Empirical Designs

The empirical work begins with information quantities and holder quantities. It does not infer distrust from spreads. This order is required by Proposition 1 and prevents the dependent variable from being used to construct the principal regressor.

Revision histories. Let y_{cvt} be a real-time official forecast for country c, vintage v, and target date t. Put every forecast in the liquid-capacity units of the model using a predetermined mapping from debt service, fiscal balances, and reserves. The revision $e_{cvt} = y_{cvt} - y_{cFt}$ compares a real-time release with the final vintage. The upper tail of positive e_{cvt} disciplines the pessimistic-bias radius. The benchmark uses a pre-specified upper quantile after subtracting estimated sampling noise. The quantile and the vintage window are fixed before any spread is used. The IMF historical WEO archive supplies public macroeconomic vintages (International Monetary Fund, 2026). Country statistical releases and Eurostat vintages can extend the fiscal block.

Signal precision. Individual forecast panels identify the information variances. In release round r, forecaster j reports x_{jcr} before observing the next official release. Country by round effects absorb the common information set. The residual variance of x_{jcr}, corrected for persistent forecaster effects and forecast-horizon effects, estimates σ_x^2 in capacity units. Real-time official forecast errors estimate σ_y^2. The public ECB Survey of Professional Forecasters microdata provide individual euro-area forecasts and revision histories (European Central Bank, 2026b). Licensed Consensus Economics forecasts extend country coverage. The two sources discipline $\beta = \sigma_x^{-2}$ and $\alpha = \sigma_y^{-2}$ without using sovereign prices.

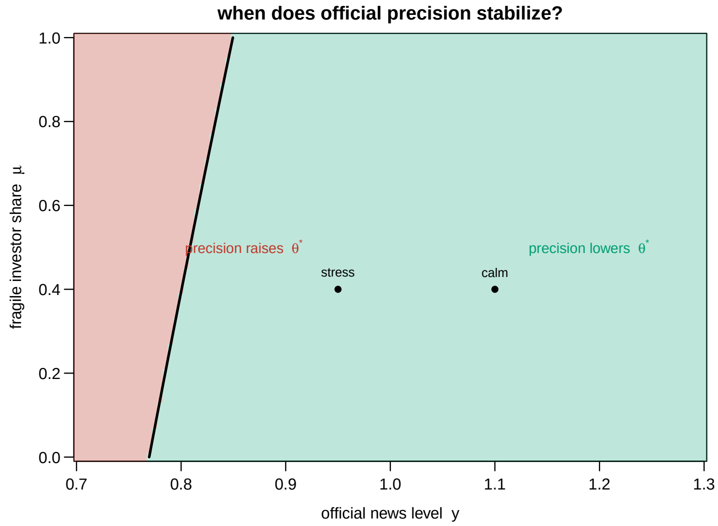

Figure 2: The sign of $d\theta^*/d\alpha$: official precision stabilizes in the green region and destabilizes in the red, separated by the boundary $y^\dagger(\mu)$ of Proposition 5. Dots mark the calm and stress calibrations at $\mu = 0.40$.

Investor composition. The empirical share μ_{ct} is the fraction of maturing marketable debt held by sectors whose mandate and funding structure make exit responsive to prices. Current euro-area sector holdings come from the ECB Securities Holdings Statistics by Sector (European Central Bank, 2026a). World Bank Quarterly Public Sector Debt and Quarterly External Debt Statistics provide current cross-country debt stocks and creditor splits where available (World Bank, 2026b; World Bank, 2026a). The panel of Arslanalp and Tsuda (2014) is used for its historical period. It is not treated as a current holder census.

Market test. With $(\delta, \alpha, \beta, \mu)$ measured before spreads, the quarterly specification is

$$\Delta s_{i,t} = \alpha_i + \tau_t + \beta_1 \, D_{i,t} \times \mu_{i,t} + \beta_2 \, D_{i,t} \times \mu_{i,t} \times \text{Stress}_{i,t} + \beta_3 \, G_{i,t} \times \mu_{i,t} + \Gamma' X_{i,t} + \varepsilon_{i,t}, \quad (6)$$

where D is the revision-based radius, G is forecast disagreement, and stress is constructed from information dated before the spread change. The theory signs β_1 when the fragile type has the higher run propensity. It predicts a larger interaction in observations where the relevant normal density is high. Proposition 6 concerns the removal of public precision, not every local variance change. The regression therefore estimates separate coefficients on revision-based distrust and forecast variance without imposing opposite signs mechanically.

Primary auctions provide the quantity counterpart. Bid-to-cover, allotment tails, and cancellations are projected on the same predetermined interaction with issue fixed effects and narrow announcement windows. These outcomes reveal rollover pressure more directly than secondary-market spreads. Public release dates also permit tests of the sign boundary in Proposition 5.

Institutional reforms supply useful timing only under explicit assumptions. Subscription

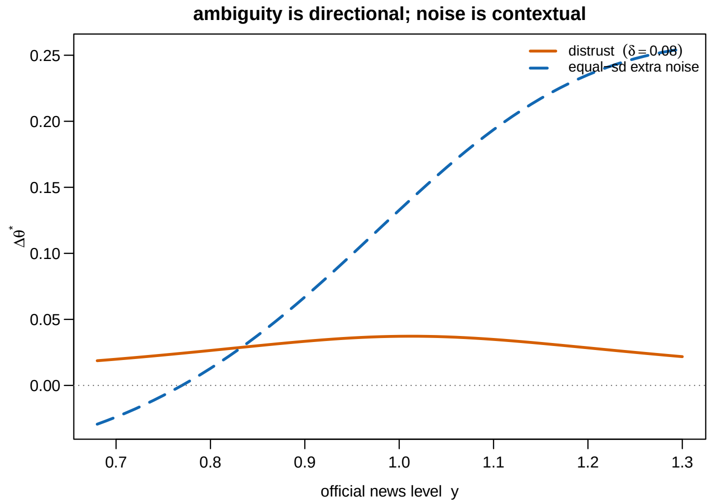

Figure 3: Finite scenario comparison. The solid line gives the threshold effect of $\delta = 0.08$. The dashed line gives the effect of raising the official signal standard deviation by 0.08.

Table 3: Predictions and implementation.

Prediction	Model magnitude	Null	Estimation
Revision-based pessimism interacts with fragile holdings	18 to 98 bp scenarios	additive effects	spread panel
The cutoff response scales with official-information dependence	loading α/β	uniform	forecast microdata
Composition shifts raise sensitivity to adverse news	10 calm, 44 stress	state invariant	holder-sector panel
Precision changes sign at $y^\dagger$, credibility does not	Figure 2	uniform sign	release windows
Complete information loss converges to the weighted payoff anchor	Proposition 6	current threshold	vintage comparisons
Effective pessimism widens forecast disagreement	Remark 1	common response	individual forecasts

to dissemination standards and adoption of fiscal councils are chosen in response to country conditions. Article IV and program publications are also selected by surveillance and program calendars. Event studies around these dates are descriptive unless reform timing is conditionally independent of latent credit news. A causal design requires a documented external eligibility rule, a staggered mandate whose timing was fixed before market conditions, or an instrument that shifts information quality without directly shifting repayment capacity. Pre-trend tests and stacked event windows cannot replace that exclusion restriction.

Table 3 states model restrictions and the data moment that bears on each. Forecast polarization cannot by itself distinguish ambiguity from heterogeneous Bayesian bias beliefs. Its role is to estimate effective pessimism. The preference interpretation requires evidence on choice under known bias distributions that lies outside the sovereign spread equation.

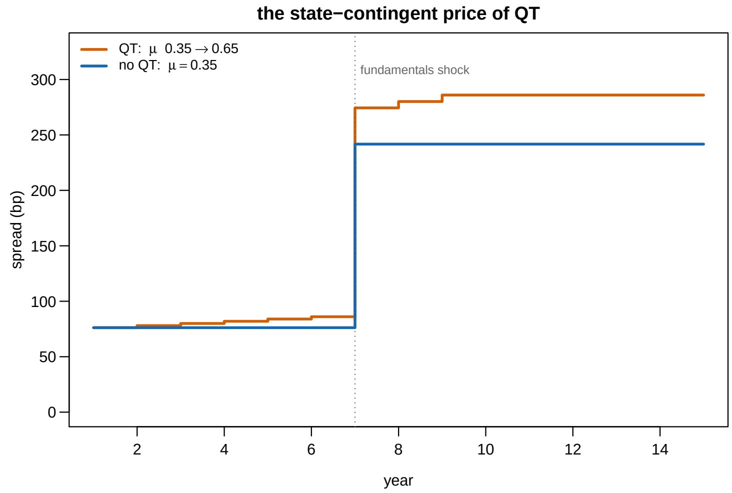

Figure 4: The state-contingent price of quantitative tightening: spread paths as the fragile share drifts from 0.35 to 0.65, against a fixed-share counterfactual. A moderate fundamentals shock arrives in year seven.

8 Conclusion

An unknown bias in official information enters the rollover game as a type-specific downward shift in the public signal. The shift raises the run cutoff by $(\alpha/\beta)\delta_k$. It leaves the global-game slope bound unchanged. Investor composition is therefore priced through a unique threshold and a transparent coordination multiplier.

The as-if result imposes an equally important limit. Equilibrium choices and prices cannot distinguish max-min ambiguity from Bayesian pessimistic bias beliefs. Revision histories must discipline the admissible bias before spreads enter the analysis. Individual forecasts identify information variances and the distribution of effective pessimism. Current holder-sector accounts identify composition. Market data then test the coordination interaction. They do not identify ambiguity preferences by inversion.

Official precision has a conditional threshold effect. Official credibility has a pointwise stabilizing effect. Ex ante welfare also depends on the distribution of announcements and on the cost of information production. The quantitative results retain this distinction. The reported 18 and 98 basis point increments describe a common-payoff scenario under calm and stress news. They establish the state dependence created by the run margin. They do not measure the ambiguity component of observed sovereign spreads.

Table 4: Data and implementation.

Dataset	Coverage	Freq.	Access	Status	Use
ECB SHSS by sector (European Central Bank, 2026a)	euro-area securities	quarterly	public aggregates	observed	current holder shares
World Bank QPSD and QEDS (World Bank, 2026b; World Bank, 2026a)	reporting economies	quarterly	public	observed	current debt stocks
Historical investor base (Arslanalp and Tsuda, 2014)	AE and EM sample	quarterly	public	observed	historical panel
Real-time official forecasts (International Monetary Fund, 2026)	WEO reporters	2/y	public	observed	revision histories
Data-standard subscriptions	IMF dissemination standards	event	public	observed	event panel
ECB SPF microdata (European Central Bank, 2026b)	euro area	quarterly	public	individual	precision moments
Consensus forecasts	country sample	monthly	licensed	individual	precision moments
Bond and CDS spreads	liquid sovereigns	daily	licensed	observed	licensed panel
Auction microdata	debt-office releases	per auction	uneven	observed where released	debt-office sample
Institutional and release dates	official archives	event	public	observed	timing tests

A Proofs

A.1 Proof of Lemma 1

Fix a profile of cutoff strategies. Its run mass is continuous and strictly decreasing in θ, while liquid capacity is strictly increasing. There is one crossing θ^* and crisis occurs exactly when $\theta \leq \theta^*$. Under bias model b, let $P_b = \mathbb{P}_b(\theta \leq \theta^* \mid x_i, y)$. The posterior mean is $[\alpha(y-b)+\beta x_i]/(\alpha+\beta)$, so P_b is increasing in b. The expected payoffs are

$$U_k^R(b) = R_k - P_b(R_k - \mathcal{R}_k), \qquad U_k^E(b) = Q_k - P_b(Q_k - q_k).$$

Assumption 1 makes both functions weakly decreasing in P_b. Their minima are therefore attained at $b = \delta_k$. Comparing the two minima gives

$$U_k^R(\delta_k) - U_k^E(\delta_k) = R_k - Q_k - P_{\delta_k}\left[(R_k - \mathcal{R}_k) - (Q_k - q_k)\right].$$

The creditor rolls precisely when $P_{\delta_k} \leq \bar{p}_k$. This is the Bayesian decision computed from effective signal $y - \delta_k$. ■

A.2 Proof of Proposition 1

Lemma 1 maps every type's max-min problem into a Bayesian problem with the same payoff hurdle and the effective public signal $y - \delta_k$. Fix any cutoff profile. The mapping gives the same action for every private signal, hence the same aggregate run function. Its crossing with liquid capacity is the same. Applying the argument to the equilibrium cutoff profile proves equality of equilibrium choices and crisis probabilities. Any price that is a fixed function of the equilibrium crisis probability and state-contingent payoffs is also the same. ■

A.3 Proof of Proposition 2

With posterior $\theta \mid x \sim N(m_k(x), (\alpha+\beta)^{-1})$, indifference requires

$$m_k(x_k^*) = \theta^* - \frac{\Phi^{-1}(\bar{p}_k)}{\sqrt{\alpha+\beta}}.$$

Since $m_k(x) = [\alpha(y-\delta_k)+\beta x]/(\alpha+\beta)$, solving for the cutoff gives $x_k^* = x_{B,k}^* + (\alpha/\beta)\delta_k$, where $x_{B,k}^*$ is the cutoff at $\delta_k = 0$ with the same payoff hurdle. Multiplication by $\sqrt{\beta}$ gives the shift $(\alpha/\sqrt{\beta})\delta_k$ in the run index. ■

A.4 Proof of Proposition 3

Substituting the cutoffs into the crossing condition gives (4). Its right side has derivative

$$\frac{\alpha}{\sqrt{\beta}} \sum_k \mu_k \phi\left(g_k + \frac{\alpha}{\sqrt{\beta}}\delta_k\right) \leq \frac{\alpha}{\sqrt{2\pi\beta}} < 1.$$

The bound is uniform in payoffs, composition, and distrust. The map sends $[0,1]$ into $(0,1)$ and is a contraction, so it has one fixed point. Each type's posterior crisis probability is strictly decreasing in its private signal, which gives one cutoff for that threshold. The resulting profile is the unique monotone equilibrium. ■

A.5 Proof of Proposition 4

Let $s = \alpha/\sqrt{\beta}$, $A_F(\theta) = \Phi(g_F(\theta) + s\delta_F)$, and $A_S(\theta) = \Phi(g_S(\theta) + s\delta_S)$. With two types, the fixed point is

$$H(\theta, \mu) = (1-\mu)A_S(\theta) + \mu A_F(\theta) - \theta = 0.$$

Write $\Phi' = s[(1-\mu)\phi_S + \mu\phi_F]$. The uniqueness condition gives $H_\theta = \Phi' - 1 < 0$. Implicit differentiation gives

$$\frac{d\theta^*}{d\mu} = -\frac{H_\mu}{H_\theta} = \frac{A_F(\theta^*) - A_S(\theta^*)}{1 - \Phi'}.$$

Its sign is the sign of the difference in the two run probabilities. Holding $\bar{p}_F$ fixed, differentiation with respect to δ_F gives

$$\frac{d\theta^*}{d\delta_F} = \frac{\mu s \phi_F}{1 - \Phi'} > 0.$$

The numerator densities locate the marginal creditors. The denominator is the coordination multiplier. Both vary with the state. ■

A.6 Proof of Proposition 5

Differentiate (4) in α. Holding θ^* fixed gives

$$\frac{\partial}{\partial \alpha}\left(g_k + \frac{\alpha}{\sqrt{\beta}}\delta_k\right) = \frac{(\theta^* - y) - \Phi^{-1}(\bar{p}_k)/(2\sqrt{\alpha+\beta}) + \delta_k}{\sqrt{\beta}}.$$

Implicit differentiation therefore gives

$$\frac{d\theta^*}{d\alpha} = \frac{\sum_k \mu_k \phi_k \left[(\theta^* - y) - \Phi^{-1}(\bar{p}_k)/(2\sqrt{\alpha+\beta}) + \delta_k\right]}{\sqrt{\beta}(1-\Phi')},$$

whose sign is that of $y^\dagger - y$. If every $\bar{p}_k < 1/2$, every quantile term is positive. The distrust terms are nonnegative. Hence $y^\dagger > \theta^*$. Direct differentiation in δ_k gives part (ii). ■

A.7 Proof of Proposition 6

Part (i) follows from direct differentiation of (4). As $\alpha \to 0$, $g_k \to -\Phi^{-1}(\bar{p}_k)$ and the distrust shifts vanish. The right side converges uniformly on $[0,1]$ to

$$\sum_k \mu_k \Phi[-\Phi^{-1}(\bar{p}_k)] = \sum_k \mu_k (1 - \bar{p}_k).$$

Uniform convergence and uniqueness imply convergence of the fixed point to this constant. Subtracting the current threshold gives the finite effect of removing public precision. Part (iii) compares that signed difference with the strictly positive derivative in part (i). ■

B Derivations: Pricing and Pass-Through

Spreads. With risk-neutral pricing at a zero reference rate, a bond promising R with crisis probability P and recovery $\mathcal{R}$ carries spread $s = P(1-\mathcal{R})/[1-P(1-\mathcal{R})]$ per period, the mapping used throughout. For small P it is $P\times$ loss given default. Endogenizing R through an issuance condition adds a second fixed point that raises all fragility statics (a higher threshold raises the promised yield, which through $\bar{p}$ raises the threshold again). The reported magnitudes are conservative in omitting it.

News pass-through and the multiplier. Differentiating (4) in y gives $d\theta^*/dy = -\Phi'/(1-\Phi')$ and hence

$$\frac{dP}{dy} = \phi\big((\theta^* - y)\sqrt{\alpha}\big)\sqrt{\alpha}\left(\frac{d\theta^*}{dy} - 1\right) = -\frac{\phi\big((\theta^* - y)\sqrt{\alpha}\big)\sqrt{\alpha}}{1 - \Phi'} \ :$$

the coordination multiplier scales the pass-through of public news to crisis risk. The direct density and the fixed-point slope must be disciplined by the forecast moments before this expression is taken to market data.

C Dynamics and the Doom Loop

Composition evolves by accounting: with runoff fraction q_t of the official portfolio's maturing stock, $\mu_{t+1} = \mu_t + q_t(1-\mu_t)\omega_t$, where ω_t is the official share of maturing debt. The QT experiment in the text uses a linear drift matching an eight-year normalization. The debt-service feedback closes fundamentals: $y_{t+1} = \bar{y}_{t+1} - \varsigma\,(s_t - s^{ref})$, service costs subtracting from next period's capacity news. Combining with the pass-through result above, the loop is locally explosive when

$$\varsigma \times \frac{\phi\big((\theta^* - y)\sqrt{\alpha}\big)\sqrt{\alpha}\,(1-\mathcal{R})}{1-\Phi'} \ > \ 1,$$

a condition that tightens mechanically as stress raises the density term and as composition and distrust raise Φ': the same statics that price spreads shrink the distance to the spiral. This is the Calvo (1988) instability with the coordination multiplier made explicit. Bocola and Dovis (2019) quantify the Bayesian analogue. The text's experiments hold $\varsigma = 0$ to isolate the state-contingent spread gap that any positive ς would amplify.

D Parameter Discipline and Sensitivity

Table 5 records where each parameter enters, its units, its source, the range examined, and what moves. At $\delta = 0$ the model is the standard Bayesian global game and the distinctive predictions

P1, P4's distrust term, P5, and P6 return to zero. As $\alpha/\sqrt{\beta}$ approaches 2.51, the multiplier grows and the distance to selection becomes the relevant statistic, with the margin discussed in Section 2.

Table 5: Parameter discipline and sensitivity.

Parameter	Enters at	Units	Source or moment	Range examined	Effect of the range
R, κ	eq. (2)	face value	gross yield, fire-sale discount	joint shift	shift $\bar{p}$ jointly
$\mathcal{R} = 0.60$	eq. (2)	recovery rate	Cruces and Trebesch (2013)	0.60 to 0.70 (CAC exp.)	stress spread 575 to 128 bp
$\bar{p} = 27.3\%$	eq. (4)	probability	implied	follows $\mathcal{R}$	anchor $1 - \bar{p}$ moves
σ_x, σ_y	eq. (4)	capacity units	scenario, forecast targets	12-cell grid, σ_x 0.10 to 0.30, σ_y 0.25 to 0.50	premia 0 to 51 / 0 to 98 bp, 3 cells cross uniqueness
$\alpha/\sqrt{\beta} = 1.35$	uniqueness	unitless	implied	up to bound 2.51	multiplier grows, then multiplicity
$\delta = 0.08$	eq. (3)	capacity units	scenario, revision-quantile target	0.04 to 0.12 (Fig. 1)	premium 18 to 98 bp across states
$\mu = 0.40$	eq. (4)	maturing-debt share	scenario, holder-sector target	0 to 1 (Fig. 1), 0.35 to 0.65 (QT)	QT premium 10 to 44 bp
y calm/stress	eq. (4)	fundamentals units	internal, calm spread 79 bp	full axis in figures	state dependence of every result

Notes: recovery is externally anchored. All other non-implied values are scenario inputs. Section 7 defines the data moments required for estimation.

E Data Appendix

Investor base. ECB Securities Holdings Statistics by Sector supply current euro-area holder aggregates (European Central Bank, 2026a). World Bank QPSD provides current public debt stocks (World Bank, 2026b). QEDS provides external debt statistics for reporting economies (World Bank, 2026a). The data of Arslanalp and Tsuda (2014) provide the historical panel covered by that study. The fragile share aggregates private sectors whose mandates make exit price-sensitive. Debt-office maturity data provide weights where they are public.

Bias-set measure. The principal measure is a pre-specified upper quantile of positive real-time forecast revisions in capacity units. The IMF historical WEO archive supplies a public vintage panel (International Monetary Fund, 2026). Data-standard subscription dates (Cady, 2004) and documented statistical-integrity events supply external validity checks. They do not enter the spread inversion because no such inversion is used.

Information precision. ECB SPF individual forecasts provide public microdata for euro-area macroeconomic expectations (European Central Bank, 2026b). Licensed Consensus forecasts extend the country panel. Within-round dispersion, persistent forecaster effects, and real-time official forecast errors identify the signal variances after all variables are mapped into common capacity units.

Market outcomes. Sovereign bond and CDS spreads at daily frequency for announcement windows and quarterly for the panel. Primary-auction results (bid-to-cover, stop-out tails, cancellations) from debt-management offices, and IMF Article IV and program-review publication dates from the IMF archive.

F Extensions

Heterogeneous private precision. If type k has private precision β_k, its cutoff shift becomes $(\alpha/\beta_k)\delta_k$ and its run index has slope $\alpha/\sqrt{\beta_k}$. A sufficient uniqueness condition is $\sum_k \mu_k \alpha/\sqrt{2\pi\beta_k} < 1$. Forecast microdata can discipline the sector-specific variances. The composition derivative retains the difference in type-specific run probabilities divided by the derivative of the aggregate fixed-point map.

Smooth ambiguity. Replacing max-min with smooth ambiguity aversion turns the worst-case shift δ_k into an endogenous pessimism weight varying with stakes. The as-if structure survives to first order, with δ_k reinterpreted as the local ambiguity premium. The max-min case is the tractable bound.

Learning about bias. Letting creditors update the bias set from realized revisions makes δ_t a state with its own persistence: credibility as reputational capital. The asymmetry between slow accumulation and one-announcement destruction suggests trap dynamics complementary to those studied here.

Dynamic runs. Staggered maturities in the sense of He and Xiong (2012) let today's rollover outcome inform tomorrow's, adding an intertemporal coordination motive to the static one. The composition state μ_t then also indexes the speed at which a run can build. The static threshold is the natural building block for that model.

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